Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels

CH Bennett, G Brassard, C Crépeau, R Jozsa, A Peres… - Physical review …, 1993 - APS

Abstract An unknown quantum state∥ φ> can be disassembled into, then later reconstructed from, purely classical information and purely nonclassical Einstein-Podolsky-Rosen (EPR) correlations. To do so the sender,"Alice,"and the receiver,"Bob,"must ...

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On the einstein-podolsky-rosen paradox

JS Bell - Physics, 1964 - World Scientific

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Quantum cryptography based on Bell's theorem

AK Ekert - Physical review letters, 1991 - APS

Abstract Practical application of the generalized Bell's theorem in the so-called key distribution process in cryptography is reported. The proposed scheme is based on the Bohm's version of the Einstein-Podolsky-Rosen gedanken experiment and Bell's theorem ...

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R Penrose - 1999 - books.google.com

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Quantum cryptography

N Gisin, G Ribordy, W Tittel, H Zbinden - Reviews of modern physics, 2002 - APS

Abstract Quantum cryptography could well be the first application of quantum mechanics at the single-quantum level. The rapid progress in both theory and experiment in recent years is reviewed, with emphasis on open questions and technological issues.

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JF Clauser, MA Horne, A Shimony, RA Holt - Physical review letters, 1969 - APS

Abstract A theorem of Bell, proving that certain predictions of quantum mechanics are inconsistent with the entire family of local hidden-variable theories, is generalized so as to apply to realizable experiments. A proposed extension of the experiment of Kocher and ...

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P Hänggi, P Talkner, M Borkovec - Reviews of Modern Physics, 1990 - APS

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Experimental quantum teleportation

D Bouwmeester, JW Pan, K Mattle, M Eibl, H Weinfurter… - Nature, 1997 - nature.com

Abstract Quantum teleportation—the transmission and reconstruction over arbitrary distances of the state of a quantum system—is demonstrated experimentally. During teleportation, an initial photon which carries the polarization that is to be transferred and ...

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Communication via one-and two-particle operators on Einstein-Podolsky-Rosen states

CH Bennett, SJ Wiesner - Physical review letters, 1992 - APS

Abstract As is well known, operations on one particle of an Einstein-Podolsky-Rosen (EPR) pair cannot influence the marginal statistics of measurements on the other particle. We characterize the set of states accessible from an initial EPR state by one-particle ...

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